PRATIK DHANAVE

AI Architect

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PROFESSIONAL SUMMARY

AI Architect with 7+ years shipping production multi-agent AI systems and cloud-native distributed backends in Python and Go across GCP, AWS, and Azure. Hands-on with LangGraph, Google ADK, Microsoft Agent Framework, MCP, RAG architecture, and vector search (pgvector, Pinecone, AlloyDB AI). Deep Telecom and Open Banking experience (UAE ADGM/DIFC, Saudi SAMA); trusted Solution Architect to 15+ enterprise clients on partner-led delivery.

ACHIEVEMENTS

- **Architecting production agentic AI for NESO** — the UK's National Energy System Operator — as AI Architect at ClearRoute (current role).
- **232 merged PRs (of 270 filed) into Microsoft's official Agent Framework for Go** — runtime correctness across the model-provider, workflow, and agent-runtime layers (highlights #472, #489); upstream merges in Google's ADK for Go and the official MCP Go SDK; core contributor to Google's Spanner Migration Tool.
- **INR 100 Cr+ saved for Tata Group** — 57% data-warehouse cost reduction via a Gemini-powered BigQuery analyzer, productized as a managed service.
- **30K+ TPS transaction engine and a 1M+ user backend** — led 10 engineers on Globe Telecom's Kubernetes platform; 47% API latency cut on the Picnic social network.
- **Speaker, Google Cloud Next 2022** — "Migrating Monolith Applications into Microservices."
- **Google Summer of Code** — Contributor (2017–2018), Mentor (2019–2026); guided 10+ students on cloud-native and AI-native projects.
- **105+ published engineering articles** across 15+ series — multi-agent systems, MCP, AI security and governance: pratikdhanave.com/blog.

CONSULTING PORTFOLIO

Industries: UAE/Saudi Open Banking · Telecom (Philippines) · FinTech · Energy · Sport · Industrial IoT · Healthcare

Engagement Types: Solution architecture · Cloud migration · FinOps · GenAI platform design · Multi-agent system delivery · Open-source upstreaming · Technical SPOC for enterprise accounts

Client Tiers: UK critical national infrastructure (NESO) · Fortune 500 (Tata Group, Globe Telecom, Dainik Bhaskar) · Tier-1 regulated banks (Standard Chartered via SC Ventures) · UAE/Saudi regulated FinTech (Bancnet — ADGM/DIFC/SAMA) · Global telco vendors (Ericsson, AT&T)

Scope: 15+ enterprise engagements across 5 countries · Case studies: pratikdhanave.com/architecture · 13 public client & colleague recommendations: pratikdhanave.com/recommendations.

TECHNICAL SKILLS

Languages: Python, Go, TypeScript, SQL
Cloud Platforms: Amazon Web Services (AWS), Google Cloud Platform (GCP), Microsoft Azure
AI Engineering: Agentic AI, Multi-Agent Systems, Agent Workflows, MLOps, LLMOps
AI Frameworks: LangGraph, LangChain, CrewAI, AutoGen, Google ADK, Microsoft Agent Framework
AI Protocols & Patterns: Model Context Protocol (MCP), A2A, Function Calling, ReAct, Chain-of-Thought, Prompt Engineering, RAG Architecture, GraphRAG, Memory Management, Embeddings, Reranking
LLM Vendors & Platforms: OpenAI, Anthropic Claude, Gemini, Vertex AI Agent Engine, AWS Bedrock Agents, Azure AI Foundry, Hugging Face, Ollama, DataRobot (familiar)
AI Quality & Ops: AI Evaluation (LLM-as-Judge), LangSmith, Guardrails, Model Routing
Vector Databases: pgvector, Pinecone, AlloyDB AI
Databases: PostgreSQL, MongoDB, Redis, Cloud Spanner, BigQuery, Azure SQL

APIs & Protocols: REST, gRPC, GraphQL, API Gateway, UPI, Stripe
Frameworks & Tools: FastAPI, Django, Gin, Node.js, Docker, Kubernetes, OpenShift (familiar), GKE, EKS, Helm, Operators, Terraform, Apache Kafka, Apache Airflow, Prometheus, Grafana, Cloud Run, AWS Lambda, Git, GitHub Actions, Jenkins, Linux
AI Dev Tools: Cursor, Claude Code, GitHub Copilot
Methodologies & Practices: Agile, Scrum, TDD, Spec-Driven Development, Code Review, CI/CD, SRE, Observability, Event-Driven Architecture, Microservices, DDD, CQRS, API-First Design
Security: Cloud IAM, Secret Manager, Cloud KMS, Cloud HSM, Cloud Armor, VPC Service Controls, IAP, Cloud DLP, Binary Authorization, Security Command Center, JWT, OAuth 2.0, OpenID Connect, Entra ID, RLS, Zero-Trust Architecture

PROFESSIONAL EXPERIENCE

AI Architect | ClearRoute *Jul 2026 – Present*

- Spearhead delivery of production agentic AI systems for NESO (National Energy System Operator) at ClearRoute, a quality engineering and cloud platforms consultancy — operating as a forward-deployed engineer between client and delivery teams.
- Design and implement multi-agent architectures on Microsoft Agent Framework, LangGraph, and MCP — translating business requirements into scalable designs within real delivery constraints.
- Led the migration of 18 production agents from Google ADK to the Microsoft Agent Framework in 90 days — documented as an 8-part engineering series.
- Lead a team of 4 Forward Deployed Engineers; set architecture standards and drive technical decision-making.

Senior Software Engineer | Searce Cosourcing Services Pvt Ltd (Google Cloud Premier Partner) *Feb 2022 – Oct 2025*

- Led the BigQuery FinOps engagement for Tata Group — built a Gemini-powered SQL anti-pattern analyzer (Optimus), refactored queries, and moved to capacity-based slot reservations, delivering **INR 100 Cr+ savings and a 57% data-warehouse cost reduction**; productized as a managed service.
- Architected the Picnic social platform backend serving **1M+ users** — Go/gRPC microservices behind a GraphQL API gateway, **reducing API response latency by 47%** via protobuf contracts and service consolidation; led the remote-first team of 4 to 80%+ automated test coverage and 40% lower cloud infrastructure costs.
- Applied AI / AI-Native Engineering team — built Python/FastAPI services and multi-agent GenAI systems (Google ADK, MCP, Microsoft Agent Framework, Vertex AI Agent Engine): RAG pipelines, Gemini analyzers, conversational assistants.
- Built and led a team of 4 engineers; mentored 40+ developers across 4 internship cohorts; set architecture standards and drove technical decision-making across product lines.
- Led technical engagement for 15+ enterprise clients as the technical SPOC — reduced onboarding time by 35% and accelerated deployments by 30%; resolved P1 incidents affecting critical client workflows.

Selected Projects at Searce:

Litmus — GCP-Powered Industrial IoT Edge Data Platform (Python, Go, MQTT, OPC-UA, GKE, Pub/Sub, BigQuery) · Senior Agentic AI Engineer · May 2025 – Oct 2025

- GCP-powered backend integrations for Litmus Industrial IoT Edge Data Platform — real-time data ingestion from manufacturing and energy plants (MQTT, OPC-UA), Python streaming pipelines to BigQuery, Vertex AI-driven anomaly detection, and edge-to-cloud orchestration on GKE + Pub/Sub.

Bancnet — GCP Agent-Powered Open Banking Portal (Python, Go, Vertex AI Agent Engine, AlloyDB AI, FastAPI, RAG) · Senior Agentic AI Engineer · Aug 2024 – Apr 2025

- GCP-agent-driven API-first Open Banking platform for UAE (ADGM/DIFC) and Saudi (SAMA) — Vertex AI Agent Engine for consent management, data residency, RAG semantic search on AlloyDB AI vector database (37% latency reduction).

Brownlow — AFL Voting Platform (Go, Python, GraphQL, gRPC) · Lead Backend Engineer · Mar 2023 – Aug 2023

- Zero-trust voting platform for AFL Brownlow Medal — 100K+ votes, 10K+ concurrent users on Cloud Run; validation, encryption (Cloud KMS), audit logging, abuse prevention via Security Command Center.

HarbourBridge — Spanner Migration Tool (Go, Python, Cloud Spanner) · Core Contributor · Mar 2022 – Aug 2022

- Core contributor to Google's open-source Spanner migration tool — backend APIs for Intelligent Schema Assistant, CDC pipelines (Datastream/Pub/Sub/Dataflow), 40–60% post-migration query gains via PK design + indexing.

Globe — Telecom/FinTech Transaction Engine (Go, Python, Kafka, Kubernetes) · Architect & Tech Lead (10 engineers) · Apr 2022 – Sep 2022

- Architected a Kubernetes transaction platform (Go, Kafka, Pub/Sub, Redis) handling **30K+ TPS** for Telecom/FinTech; **led 10 engineers**; idempotent processing, ledger/reconciliation, dual-layer auth (JWT + API Keys), PCI-aligned data protection, error-code orchestration with exponential backoff and DLQ fallback.

Specialist Application Developer | Net Connect Pvt Ltd *Jun 2021 – Feb 2022*

Selected Project at Net Connect:

azure-service-operator — Kubernetes-Native Cloud Lifecycle for Ericsson & AT&T (Go, Python, Azure, Kubernetes) · Open Source Developer & Architect · Jun 2021 – Feb 2022

- Served as Open Source Developer and Architect for airshipit and azure-service-operator in collaboration with Ericsson, AT&T, and Microsoft.
- Integrated OpenTelemetry observability framework into the airshipit project, reducing manual operations by ~30% and improving incident detection and recovery time.

Software Developer | CloudCover Pvt Ltd (Google Cloud Premier Partner) *Oct 2019 – May 2020*

Selected Project at CloudCover:

Bloom (SC Ventures) — Secure Cloud Provisioning for Standard Chartered (Terraform, AWS, Azure, SOC 2, ISO 27001) · Cloud Engineer · Oct 2019 – May 2020

- Contributed to Bloom (backed by SC Ventures, Standard Chartered) — built Terraform-based, policy-driven secure cloud provisioning for regulated financial institutions on AWS/Azure with SOC 2 and ISO 27001 controls; developed push-to-deploy self-service capabilities, cutting onboarding time by 20%.
- Designed multi-cloud landing-zone patterns for regulated banking workloads; introduced policy-as-code guardrails (Open Policy Agent) and centralized audit logging.

Software Developer | SoftCorner *Feb 2016 – Dec 2017*

Selected Project at SoftCorner:

Shelter Associates — Open-Source Data Platform for Slum Mapping (Python, KoBoToolbox, Google Maps, BIRT) · Full-Stack Developer · Feb 2016 – Dec 2017

- Built an open-source data platform for Shelter Associates — integrated KoBoToolbox for survey data ingestion, built Python data pipelines, automated reporting via BIRT, and delivered spatial data visualization on Google Maps.
- Supported NGO-led slum mapping and rehousing programs across Pune, Sangli, and Kolhapur — delivered geospatial dashboards that informed municipal policy decisions and household-level housing interventions.

OPEN SOURCE CONTRIBUTIONS

- Gocloud: Developed a unified API library for managing cloud resources across AWS, GCP, and Azure.
- CX Programming Language: Enhanced CX interpreter for deterministic blockchain transaction execution; implemented Pratt parser reducing parsing complexity by ~30%.
- Spanner Migration Tool: Redesigned schemas to prevent write hotspots; built CDC pipelines (Datastream, Pub/Sub, Dataflow) enabling minimal-downtime migrations to Cloud Spanner.
- Microsoft Agent Framework Go: 232 merged PRs (of 270 filed) across the model-provider, workflow, and agent-runtime layers, each with race + regression tests — highlights: tool-approval nil-panic hardening (#472), route-builder handler validation (#489).

- Google ADK (Go): 12 merged PRs (of 18 filed) into Google's official Agent Development Kit for Go — runtime-correctness fixes including a streaming-aggregator nil guard (#1802) and MaxConcurrentRuns enforcement in the Pub/Sub controller (#1720).
- MCP Go SDK: 8 merged PRs into the official Go SDK for the Model Context Protocol — fixed a MemoryEventStore panic on out-of-range indexes (#1132) and JSON-RPC response summaries (#1382).

PERSONAL PROJECTS

Genie — Azure AI Foundry-Driven Multi-Agent Financial Assistant (Python, Go, Azure AI Foundry, Microsoft Agent Framework, Postgres, OpenTelemetry) · Author & Lead Engineer · Nov 2025 – Jun 2026

- 15 role-specialized agents (supervisor, analyzer, forecaster, anomaly_detector, recommender, llm_auditor, etc.) on Azure AI Foundry + Microsoft Agent Framework; JWT + RBAC, AES-256-GCM envelope encryption, end-to-end OpenTelemetry tracing across agent hops.

Bodh — Microsoft-Powered MAI-DxO-Style Medical Diagnostic Agent Panel (Python, Go, Microsoft Agent Framework, FHIR R4, SD-Bench) · Author & Lead Engineer · Nov 2025 – Jun 2026

- Virtual physician panel with role-specialized agents (intake, supervisor, questioner, test_planner, cost_guardian, diagnostician, reasoning_verifier); FHIR R4 + HL7 v2 aware; cost-aware diagnostic budget enforcement.

CERTIFICATIONS

- LinkedIn Learning: Hands-On AI: Building Agents with the Google Agent Development Toolkit (Google ADK) — Jun 2026.
- Udemy: AWS Certified Generative AI Developer – Professional.
- Coursera: Google Cloud — Generative AI Leader Professional Certificate.
- Ardan Labs: Ultimate Go and Ultimate Services Go.

EDUCATION

Bachelor of Engineering in Information Technology *2015*

Pune Institute of Computer Technology, University of Pune, India

LANGUAGES

English (Professional — full business fluency, written and spoken) · Hindi (Native) · Marathi (Native)

RELOCATION & AVAILABILITY

- **Open to India (Pune, Bengaluru, Hyderabad, Mumbai, Chennai) and international relocation — Dubai/UAE, UK, Ireland, Netherlands, Germany, Switzerland, Norway, Sweden, Denmark, Finland, Canada, USA, Singapore, New Zealand.** Passport valid through 2028, ready for work-visa sponsorship. Available within 30 days — documents ready for attestation.